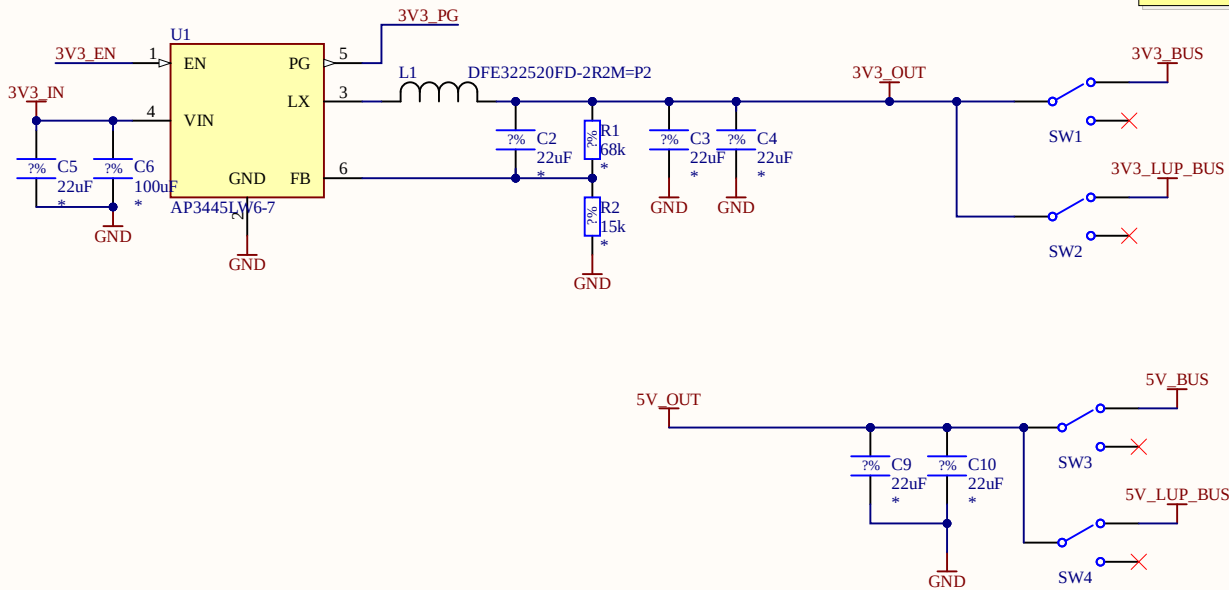


Cleanroom PSM

3V3 Regulator

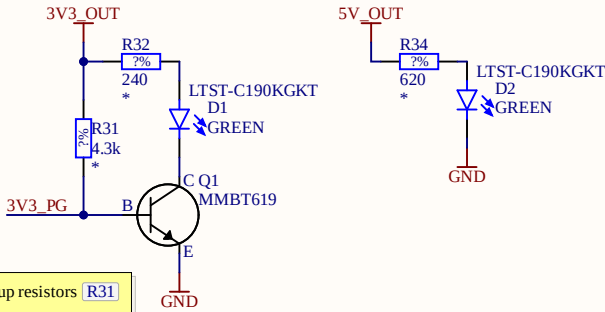


PG is an open drain output and an active high signal.
If power is good, the base of the transistor will be pulled high and the LED will turn on.

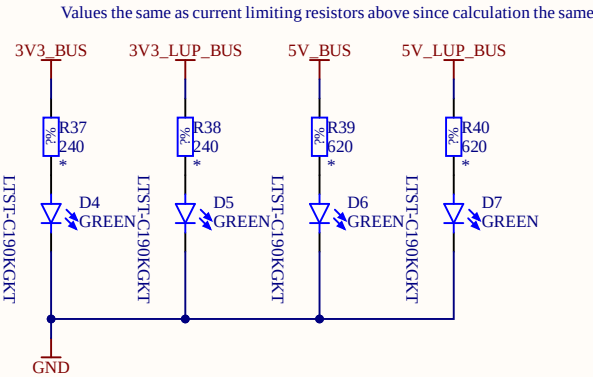
To select the current-limiting resistors, I used
 $R = V/I = (V_{rail} - V_{diode} - V_{ce(sat)})/i_{LED}$
I selected i_{LED} as 5mA for a luminosity of 4.5 to 18 mcd.
 $V_{diode} = 2.15V$
 $V_{ce(sat)} = 0V$ (worst-case assumption, in practice more like 200mV)

There are two constraints on the pull-up resistors R_{31} , R_{31} , and R_{31} .
Firstly, they must be sufficiently small that when PG is floating (logic 1) enough current can flow through to bias the transistor in saturation. We then choose $\beta = 10$ for safety, meaning that for the 5mA i_c we need $I_b = 0.5mA$, and then
 $R = V/I = (V_{cc} - V_{be(sat)})/I_b$
This immediately leads to a problem:
 $R_{31} : < 5.2k$, $R_{31} : < 8.6k$, $R_{31} : < 14.6k$
Secondly, they must be sufficiently large that when PG is pulled down to ground it does not sink more than 1mA into PG. $R = V/I = V_{cc}/1mA$
 $R_{31} : > 3.3k$, $R_{31} : > 5k$, $R_{31} : > 8k$
Taking the mean of each of these ranges and rounding to the nearest E24,
 $R_{31} : 4.3k$, $R_{31} : 6.8k$, $R_{31} : 11k$

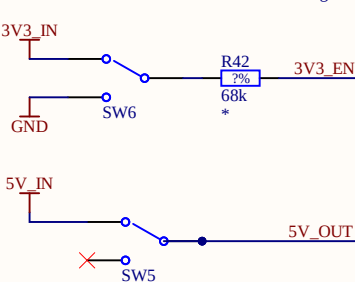
Power Good LEDs



Bus Live LEDs



Regulator Enable Switches



LEDs removed as not required with power good lights.

EN pin contains internal 2.8V zener diode and 35K resistor to GND.

RIP: The original three regulators

Title: Cleanroom PSM	
Design Name: Cleanroom PSM	Revision: V1
Project: Cleanroom	Designer: Jasper M-W
Date: 5/06/2026	File: Overview.SchDoc
Time: 9:40:44 pm	Document Size: A3



A

B

C

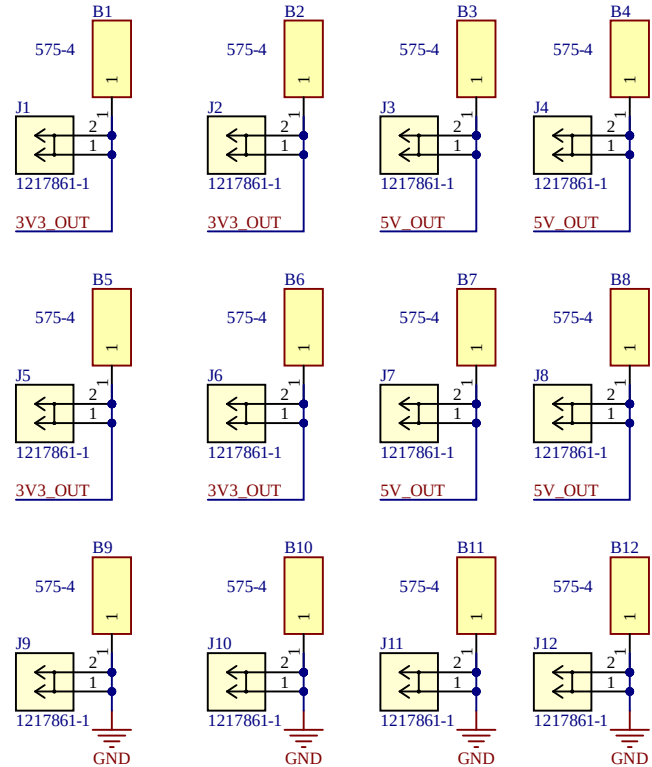
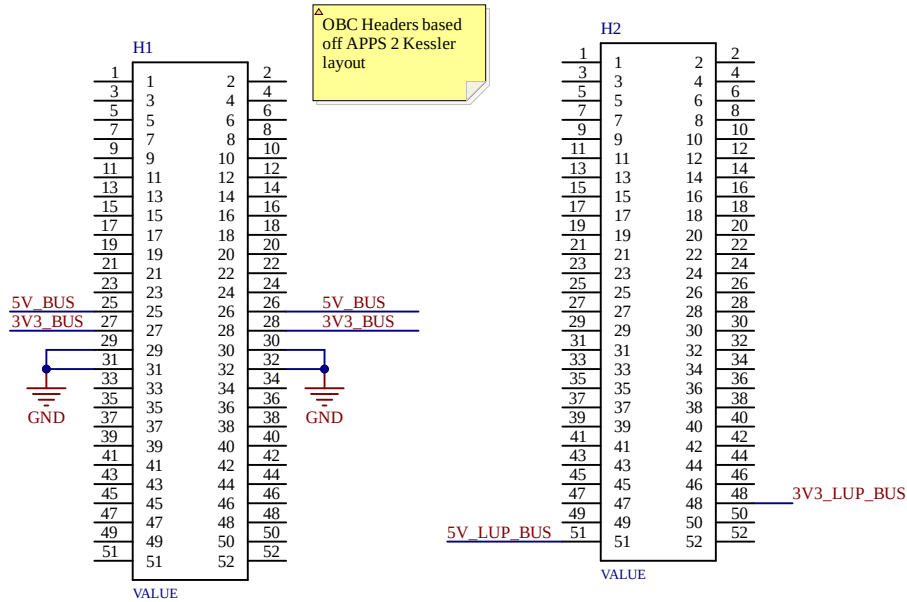
D

A

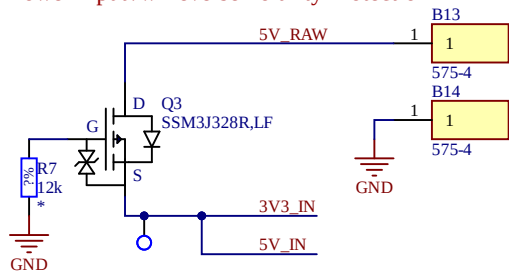
B

C

D



Power Input /w Reverse Polarity Protection



Title: PSM Interfaces	
Design Name: Cleanroom PSM	Revision: V1
Project: Cleanroom	Designer: Jasper M-W
Date: 5/06/2026	File: Interfaces.SchDoc
Time: 9:40:44 pm	Document Size: A3

